

Problem 16 — Moon-Reflected Light & Photoelectric Effect

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Problem 16 — Photon intensity, photon flux, photocurrent, work function, stopping potential. Companion piece to today's Lecture Note (Problems 7–15).

⚡ How to use this worksheet

Six parts (a → f) all share the same Setup. Solve each one in the dashed work-space first — *don't peek at the green Solution box until you've tried*. Final answers are boxed in green. Notice how part (a) → (b) → (c) → (d) is one continuous chain: each answer feeds the next.

SETUP

The average earth–moon distance is $3.84 \times 10^8 \text{ m}$. Suppose the moon reflects light with an average wavelength of 550 nm . Suppose the intensity of this radiation on a given clear night, neglecting atmospheric losses, is 0.10 W/m^2 .

Given: $d = 3.84 \times 10^8 \text{ m}$, $\lambda = 550 \text{ nm}$, $I = 0.10 \text{ W/m}^2$

USEFUL CONSTANTS

$h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$ · $c = 3.00 \times 10^8 \text{ m/s}$ · $e = 1.602 \times 10^{-19} \text{ C}$ · $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ ·
 $hc \approx 1240 \text{ eV}\cdot\text{nm}$

Part (a) Average photon energy

QUESTION

What is the average energy of the photons reflected by the moon?

WORK SPACE — YOUR SOLUTION

SOLUTION

Formula. $E = hf = \frac{hc}{\lambda}$. Use the shortcut $hc = 1240 \text{ eV}\cdot\text{nm}$.

Substitute. $E = \frac{1240 \text{ eV}\cdot\text{nm}}{550 \text{ nm}} \approx 2.255 \text{ eV}$.

In joules. $E \approx 2.255 \times 1.602 \times 10^{-19} \approx 3.61 \times 10^{-19} \text{ J}$.

$$E \approx 2.26 \text{ eV} \approx 3.61 \times 10^{-19} \text{ J}$$

Part (b) Photons striking 1 m^2 on earth per second

QUESTION

How many photons strike a square meter on earth per second?

WORK SPACE — YOUR SOLUTION

SOLUTION

Idea. Intensity = energy flux. Divide by energy-per-photon to get photon flux.

$$\frac{N}{A \cdot t} = \frac{I}{E_{\text{photon}}}.$$

Substitute. $\frac{N}{(1 \text{ m}^2)(1 \text{ s})} = \frac{0.10 \text{ W/m}^2}{3.61 \times 10^{-19} \text{ J}} \approx 2.77 \times 10^{17} \text{ photons/m}^2/\text{s}.$

$$N/(\text{m}^2 \cdot \text{s}) \approx 2.77 \times 10^{17} \text{ photons}$$

Scale check. A typical clear night $\rightarrow$ about 10^{17} visible photons hitting every square meter per second from the moon alone. Light is "lumpy," but the lumps are absurdly numerous.

Part (c) **Photons striking a 1 cm^2 photocell per second**

QUESTION

How many photons strike a photocell, **1 cm** on a side, per second?

WORK SPACE — YOUR SOLUTION

SOLUTION

Convert area. $A = 1\text{ cm}^2 = (10^{-2}\text{ m})^2 = 10^{-4}\text{ m}^2$.

Scale (b) by area. $N_{\text{cell}}/s = (2.77 \times 10^{17}) \times 10^{-4} \approx 2.77 \times 10^{13}\text{ photons/s}$.

$$N_{\text{cell}} \approx 2.77 \times 10^{13} \text{ photons/s}$$

Part (d) Ideal-photocell photocurrent

QUESTION

If the photocell is ideal (each photon causes one electron to be emitted), what is the photocurrent?

WORK SPACE — YOUR SOLUTION

SOLUTION

Idea. 1 photon $\rightarrow$ 1 electron, so the number of electrons per second equals the number of photons per second. Each electron carries charge e :

$$I_{\text{cur}} = \frac{N_{\text{cell}}}{t} \cdot e.$$

Substitute. $I_{\text{cur}} = (2.77 \times 10^{13}) \times (1.602 \times 10^{-19}) \approx 4.44 \times 10^{-6} \text{ A}.$

$$I_{\text{cur}} \approx 4.44 \mu\text{A}$$

Unit sanity. photons/s $\times$ C/photon = C/s = A. Makes sense.

Part (e) Maximum allowable work function

QUESTION

What is the maximum work function of the cell to emit this current?

WORK SPACE — YOUR SOLUTION

SOLUTION

Idea. For photoemission to occur, the photon must carry at least the work function's worth of energy: $hf \geq W_0$. The *maximum* allowable W_0 is the case of equality — i.e., W_0 exactly equals the photon energy.

Plug in (a). $W_{0,\max} = E_{\text{photon}} \approx 2.26 \text{ eV} \approx 3.61 \times 10^{-19} \text{ J}$.

$$W_{0,\max} \approx 2.26 \text{ eV} \approx 3.61 \times 10^{-19} \text{ J}$$

What if W_0 is bigger? The photon would not have enough energy; no electrons ejected; $I_{\text{cur}} = 0$.

Part (f) Stopping voltage when $W = 1.0 \text{ eV}$

QUESTION

If the work function is 1.0 eV , what is the stopping voltage of the emitted electrons?

WORK SPACE — YOUR SOLUTION

SOLUTION

Einstein's equation. $KE_{\text{max}} = hf - W_0 = E_{\text{photon}} - W_0$.

Substitute. $KE_{\text{max}} = 2.255 \text{ eV} - 1.0 \text{ eV} = 1.255 \text{ eV}$.

Stopping potential. $eV_s = KE_{\text{max}} \implies V_s = \frac{KE_{\text{max}}}{e}$. Numerically (in eV): V_s in volts equals KE_{max} in eV.

$$V_s \approx 1.26 \text{ V}$$

Connection to the lab. If you ran an I–V sweep on this photocell, you would find the photocurrent drops to zero at $V \approx -1.26 \text{ V}$. That cut-off point is the stopping voltage — it's how V_s is measured experimentally.